System Requirements Specification

Memphis Music Initiative - Data Management System

1. Introduction.....1
1.1. Purpose..... 1
1.2. Product Scope..... 1
1.3. Target Users..... 2
1.4. Definitions and Acronyms.....2
2. Overall Description.....2
2.1. Overview.....2
2.2. Key Features..... 3
2.2.1. Program Structuring.....3
2.2.2. Activity and Session Scheduling.....3
2.2.3. Data Collection.....3
2.2.4. Grant Reporting and Data Dashboards..... 3
2.2.5. Data Import and Export.....3
2.2.6. Accessibility..... 3
2.2.7. Database Management.....3
2.2.8. Deployment.....4
2.2.9. Implementation Details..... 4
2.3. Technology Stack.....4
3. Features and Requirements..... 5

1. Introduction

1.1. Purpose

This document serves as a comprehensive guide to the requirements of the Memphis Music Initiative Data Management System (hereafter, “DMS”), detailing its functional and non-functional aspects.
The purpose of this project is to develop and implement a software solution that will efficiently collect, manage, and analyze information from students enrolled in partner programs of the Memphis Music Initiative (MMI). The DMS will serve as a central platform for program management, enrollment tracking, and progress monitoring. It will also facilitate the collection of surveys from program participants, with the primary goal of using the collected data for grant reporting and providing valuable insights to MMI and its partner organizations.

1.2. Product Scope

The key objectives of the Data Management System (DMS) project include:

- **Data Collection**: Develop a user-friendly interface for collecting data from students participating in MMI partner programs. This includes demographic information, program enrollment details, and survey responses.
- **Program Management**: Implement features for MMI and its partner organizations to manage program details, including program schedules, participant lists, and program-specific data.
- **Data Analysis**: Create tools and reports for analyzing and visualizing collected data to provide valuable insights into program effectiveness and participant outcomes.
- **Grant Reporting**: Ensure that the DMS can generate the necessary reports and data summaries required for grant reporting purposes, in compliance with MMI and partner organizations' reporting requirements.
- **Scalability and Accessibility**: Design the system to accommodate future growth and expansion of partner programs while ensuring accessibility and usability for all authorized users.

1.3. Target Users

DMS is designed to serve the needs of various stakeholders, including:

- **Memphis Music Initiative (MMI)**: MMI staff and leadership responsible for program management and reporting.
- **Partner Organizations**: Organizations collaborating with MMI to deliver partner programs.
- **Program Administrators**: Individuals managing program details and schedules.
- **Data Analysts and Researchers**: Users leveraging DMS for data analysis and insights.

1.4. Definitions and Acronyms

Term/Acronym	Definition
MMI	Memphis Music Initiative
DMS	Data Management System
Python	A general-purpose programming language, popular for both ease-of-use and vast ecosystem of open-source code and projects.
Django	A Python web application framework (WAF), software designed to facilitate the development of web applications.

2. Overall Description

2.1. Overview

The DMS is a sophisticated software solution designed to streamline and enhance the management of partner programs and the collection of critical data related to students enrolled in these programs, activities, and activity sessions. Developed as a Python Django application, DMS offers a comprehensive set of features and functionalities to empower MMI and its partner organizations in effectively managing, analyzing, and reporting program-related data.

2.2. Key Features

2.2.1. Program Structuring

The DMS will allow users to electronically represent MMI partner programs, enabling the efficient organization of program details, objectives, and schedules.

2.2.2. Activity and Session Scheduling

With the DMS, program administrators will be able to easily schedule activities and sessions within partner programs, ensuring smooth program execution and participant engagement.

2.2.3. Data Collection

The DMS will facilitate the collection of forms and surveys related to students enrolled in programs, activities, or activity sessions. It will also support the collection of aggregate data, allowing for the recording of program-wide statistics.

2.2.4. Grant Reporting and Data Dashboards

The system will compile collected data into comprehensive grant reports, ensuring compliance with grant requirements. It will also generate data dashboards for partners, providing valuable insights into program effectiveness and participant outcomes.

2.2.5. Data Import and Export

The DMS will support the import of data from external systems via file import, allowing for seamless integration with other tools and databases. It will also provide the capability to export data into common formats such as CSVs or Excel Workbooks for further analysis and reporting.

2.2.6. Accessibility

The DMS will be designed to be accessible from both desktop and mobile devices, ensuring that users can conveniently access and interact with the system regardless of their preferred platform. The responsive design of the user interface will adapt to different screen sizes and resolutions, providing a seamless experience on desktops, laptops, tablets, and smartphones.

2.2.7. Database Management

The DMS will leverage a PostgreSQL database for data storage and management. PostgreSQL is known for its reliability and scalability, making it well-suited for handling the diverse data needs of the system.

2.2.8. Deployment

The DMS will be deployed to one or more virtual private servers (VPS). This deployment strategy ensures the system's availability, scalability, and security. It also allows for efficient maintenance and updates.

2.2.9. Implementation Details

The DMS is being developed as a Python Django application, ensuring robustness, scalability, and maintainability. It adheres to best practices in software development and database management to guarantee data accuracy and security. The project is committed to delivering a user-friendly and intuitive interface for all users, making it accessible and efficient for program management and data handling.

2.3. Technology Stack

The DMS project is built on the following technology stack:

- **Programming Language**: Python
- **Web Framework**: Django
- **Database Management System**: PostgreSQL (or as specified)
- **Front-End**: HTML, CSS, JavaScript
- **Data Export Formats**: CSV, Excel Workbook

3. Features and Requirements

ID	Group	Description	Status	Acceptance Criteria
1	Organization	The System must support an organizational heirarchy of Partners and Programs	Proposed	- The data model must include a one-to-many relationship between Partners and Programs - The data model must include a one-to-many relationship between Programs and Activities - The data model must include a one-to-may relationship between Activities and Sessions
2	Organization	Different classes of users must have different levels of access	Proposed	- The data model must include an Admin user classification for MMI Staff - The data model must include a Partner Staff user classification for Partner staff members - The data model must include a Client user classification for clients/students enrolled in Partner programs - The data model must include a Parent user classification for parents/guardians of Client users
3	Organization	Each type of user should have a home page that presents an appropriate navigation menu	Proposed	- MMI Staff home page should display a list of Partners - Partner Staff home page should display a list of Programs - Client home page should display a list of Forms that need to be completed and a list of Forms already completed - Parent home page should display a list of Forms that need to be completed and a list of Forms already completed
4	Organization	Each user's home page should allow that user to access and enter appropriate data	Proposed	- MMI Staff home page should allow accessing and viewing Partner data - Partner Staff home page should allow accessing and viewing Partner data - Client home page should allow accessing and viewing Client data

				- Parent home page should allow accessing and viewing child Client data
5	Organization	Parent users have access to their child's data and can submit data for their child	Proposed	- Parent home page should display all Forms that need to be completed and Forms already completed for their child - Parent users can submit Forms that need to be completed on behalf of their child
6	Organization	All Forms are associated with at least one Partner	Proposed	- The data model must include a one-to-many relationship between Partners and Forms - MMI Staff can create Partners as an organizational tool (without a corresponding business partner) for the sole purpose of collecting data if desired
7	Security	All users will need to sign in to the system in order to access their home page	Proposed	- Navigation guards prevent unauthorized access to any page other than the login screen
8	Security	MMI Staff administer Partner accounts	Proposed	- MMI Staff can create, edit, and delete Partner profiles - MMI Staff can view Partner home pages, dashboards, and reports - MMI Staff can create, edit, and delete Users - MMI Staff can assign Users to Partners
9	Security	MMI Staff can access all system data	Proposed	- MMI Staff can access the systems' Admin interface - MMI Staff have administrative access to the PostgreSQL database

10	Security	Each Partner Staff user must be associated with a single Partner	Proposed	- The data model must include a one-to-many relationship between Partners and Users designated as a Partner User - Partner Users may view/access all data associated with any Program associated with the same Partner as the Partner User
11	Security	Partner Staff users may not access data collected or retained by a Partner other than the one to which the user is assigned	Proposed	- The system must include navigation guards that prevent Partner Staff users from accessing any page in the application that would display data associated with a Partner other than the one to which the user is assigned
12	Security	Data changes shall be tracked in an Audit Log	Proposed	- The Audit Log must contain the name of the table containing the record, the date the data was changed, the user who made the change, the type of change (create, update, delete), and a copy of the record resulting from the data change - The Audit Log must capture data any time a record is created - The Audit Log must capture data any time a record is updated or modified - The Audit Log must capture data any time a record is deleted - The Audit Log should retain records of data changes for 90 days

13	Administration	Partner Staff may associate Forms with Programs, Activities, or Sessions to be completed by Clients	Proposed	- Partner Staff must be able to modify the associations of Forms assigned to that Partner - The data model must include an optional many-to-many relationship between Programs and Forms - The data model must include an optional many-to-many relationship between Activities and Forms - The data model must include an optional many-to-many relationship between Sessions and Forms
14	Administration	MMI Staff can create and modify Forms	Proposed	- MMI Staff can create new forms and assign those forms to Partners, Programs, Activities, Sessions, and/or Clients - MMI Staff can add new sections to existing Forms - MMI Staff can add new fields to existing Form sections - MMI Staff can remove sections from existing Forms - MMI Staff can remove fields from existing Forms
15	Administration	The System will be configured with existing Partners and Forms	Proposed	- All Partners named in the Design Document will be configured in the System at launch - All Forms submitted to the Google Drive prior to the fourth Sprint will be configured in the System at launch
16	Administration	Partner Users with elevated permissions can make changes to the structure of Forms for that Partner	Proposed	- Partner Staff can create new forms and assign those forms to Programs, Activities, Sessions, and/or Clients already associated with that Partner - Partner Staff can add new sections to existing Forms associated with that Partner - Partner Staff can add new fields to existing Form sections associated with that Partner - Partner Staff can remove sections from existing Forms associated with that Partner - Partner Staff can remove fields from existing Forms associated

				with that Partner
17	Data Collection	Data will be collected from Clients via Forms	Proposed	- Forms must present a fillable form to Clients and other Users via the System interface - Completed Forms should not be editable
18	Data Collection	Client users must have access to complete Forms or view completed Forms	Proposed	- The system must provide access to Clients to complete any Form assigned to a Program, Activity, or Session to which the Client is associated - The system must provide access to Clients to complete any Form assigned directly to that Client - Client users must have access to view any Form that Client has completed

19	Data Collection	A Form consist of one or more sections, where each section consists of a variety of Form Fields	Proposed	- The data model must include a one-to-many relationship between Forms and Form Sections - The data model must include a many-to-many relationship between Form Sections and Form Fields - The System must support Text Fields for short text entry - The System must support Text Areas for long text entry - The System must support Single Select fields for choosing a single option - The System must support Multiple Select fields for choosing multiple options - The System must support Date fields for capturing dates
20	Data Collection	The System must allow Users to access and complete Forms without an active internet connection	Proposed	- The application on a user's device must download new forms while the user has an active internet connection, to be completed at any time. - The applicaiton on a user's device must allow users to complete downloaded/cached forms - The application on a users's device must submit forms completed while the user was offline once an active internet connection has been established.
21	Data Collection	Users may upload data into the System via CSV or Excel data import	Proposed	- The System must allow Partner Users to choose a file to upload to the System - All data file templates submitted to the Google Drive prior to the fourth Sprint will have an import function defined - Data submitted via import will be stored in the data base and

				adapted to the data model
22	Data Presentation	Data Dashboards will be made available to MMI Staff	Proposed	- MMI Dashboards may include aggregate data for any and all Partners - Specific metrics to be determined
23	Data Presentation	Data Dashboards will be made available to Partner Users	Proposed	- The User must be able to select a time period for the data to be shown - The User must be able to select one or more Programs for the data to be shown - The Dashboard must include the following: - The number of clients enrolled in all programs selected during the chosen time period - The number and percent of clients who participated in at least one Activity associated with at least one selected Program during the chosen time period - A chart indicating, for each Activity during the time period, the number and percent of Clients assigned to that Activity that participated - The number of clients enrolled in all programs selected during the chosen time period, grouped by Race - The number of clients enrolled in all programs selected during the chosen time period, grouped by Gender - The number of clients enrolled in all programs selected during the chosen time period, grouped by Age - The number of clients enrolled in all programs selected during the chosen time period, grouped by Zip Code - Up to three additional metrics per Partner
24	Data Presentation	Downloadable reports will be made available to MMI Staff and	Proposed	- Reports must be downloadable in CSV or Excel format - Reports must be shareable via e-mail - Up to thirty (30) reports will be designed in partnership with MMI and Partners

		Partner Users		
25	Data Presentation	MMI Staff can make limited modifications to Dashboards within the System	Proposed	- The System will include a menu of configurable Dashboard components - MMI Staff can add a component to a Dashboard - MMI Staff can remove a component from a Dashboard - MMI Staff can update the configuration of a Dashboard component